Answer all questions, each carries 3 marks. Marks

1 Let the variable x be "dog" and the variable y be "cat". Write the values returned by the following operations: a) x*4 + ' ' + 4*y b) x*len(x+y) (3)

2 What will be the output if the following code fragments are executed? (3)

```
for j in range(2,10,4):
    print(j)
```

3 Write a python program to create a list of squares for the numbers from 0 to 9 using list comprehension. (3)

4 Write a user-defined function to generate even numbers between 1 and 25. (3)

Module I

11 a) Write a python program to find out the eldest and youngest of three individuals , with their ages being input through the keyboard.(without using max, min functions) (7)

b) Consider the mathematical expression $(a+b)^2=a^2+2ab+b^2$. Write a Python program that takes user input for values of a and b, then evaluates both sides of

the expression. Finally, compare the results and print whether the equation holds true or false.

OR

12 a) Enumerate the various selection structures and control statement types in Python and elucidate each with suitable examples. (8)

b) Write a python program to print factorial of a given number. (6)

Module II

13 a) Describe the concept of recursive function in Python with suitable example. (7)

b) Explain how to read numeric values from a file, perform some operations, and then write the results back to the file? (7)

OR

14 a) Compare and contrast the fundamental characteristics and use cases of lists, tuples, and sets in Python. (7)

b) Create a Python program that uses a dictionary to store the names and ages of people. Ask the user to enter a name, and the program should display the age of that person. (7)

Answer all questions, each carries 3 marks. Marks

1 Evaluate the following Python expressions and write the result: (3)

a) 2**3**2

b) round(83.37)

c) chr(ord('A') + 3)

2 Write a Python program to count the positive and negative numbers in a given set of numbers. (3)

3 What is meant by abstraction? How does Python's function serve as abstraction mechanisms? (3)

4 Assume that the variable *mydata* refers to the list [5, 3, 7] . Write the values of the following expressions: (3)

a) *mydata*[0:2]

b) *mydata*[-1]

c) tuple(*mydata*)

Answer one full question from each module, each carries 14 marks.

Module I

11 a) Given the income amount, write a Python program to compute the income tax as per following tax rules: (7)

1. If the income amount <= Rs. 2,50,000 then zero tax.
2. If the income amount <= Rs. 5,00,000 then tax is 5% of the amount exceeding Rs. 2,50,000
3. If the income amount <= Rs. 10,00,000 then tax is Rs. 12500 + 10% of the amount exceeding Rs. 5,00,000
4. If the income amount > Rs. 10,00,000 then tax is Rs. 62500 + 20% of total income exceeding Rs. 10,00,000

b) Write a Python program to check whether a number is prime or not. (7)

OR

OR

12 a) Write a Python program to compute the Highest Common Factor(HCF) of two numbers. (7)

b) Given the value of x, write a Python program to evaluate the following series upto n terms. (7)

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \frac{x^8}{8!} - \dots$$

Module II

13 a) Write a Python program that takes your full name as input and displays the abbreviations of the first and middle names followed by the last name which is displayed as it is. For example, if your name is Rani Bennet Rose, then the output should be R.B.Rose. (7)

b) Assume that a file contains integers separated by newlines. Write a Python code segment that opens the file and prints the average value of the integers. (7)

OR

14 a) The factorial of a positive integer n , fact(n) , is defined as follows: (7)

fact(n) = 1, when n = 1

fact(n) = n * fact(n - 1) , otherwise.

Write a Python program using function to find the factorial of a given positive integer. Also write another program using a recursive function fact, that returns the factorial of a given positive integer.

b) Assume that the variable *my_data* refers to the dictionary {'b':20, 'a':35} . Write the expressions that perform the following tasks: (7)

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1. Replace the value at the key 'b' in *my_data* with that value's negation.
2. Add the key/value pair 'c':40 to *my_data* .
3. Remove the value at key 'b' in *my_data* , safely.
4. Print the keys in *my_data* in alphabetical order.
5. Print a list of tuples containing the keys and values for each entry in *my_data*.

PART A

Answer all questions, each carries 3 marks.

Mark:

1 What is the output of the following print statement in Python? (3)

(a) print (9//2) (b) print (9/2)

2 Write a Python program to count number of even numbers and odd numbers in a given set of n numbers. (3)

3 Illustrate the use of negative indexing of list with example. (3)

4 Write Python code for the following statements (3)

i)writes the text "PROGRAMMING IN PYTHON" to a file with name code.txt

ii) then reads the text again and prints it to the screen.

Module I

11 a) Write a python program to generate the following type of pattern for the given N rows . (6)

```
1
1 2
1 2 3
1 2 3 4
```


b) Mention the different types of loop and control statements allowed in Python and explain each type with suitable examples. (8)

OR

12 a) Write the python program to print all prime numbers less than 1000. (7)

b) Write a Python program to find distance between two points (x1,y1) and (x2,y2). (7)

Module II

13 a) Write a Python program to count how many times each character appears in a given string and store the count in a dictionary with key as the character. (7)

b) Create a function min_max() that takes n numbers as list argument and return the smallest and largest numbers. (7)

OR

14 a) Write a Python program to read n integers into a list and separate the positive and negative numbers into two different lists. (7)

b) Create a dictionary of names and birthdays. Write a Python program that asks the user to enter a name, and the program display the birthday of that person. (7)

Module III

PART A

Answer all questions, each carries 3 marks. Mar

1 Explain type conversion with example. (3

2 Write a program that accepts the lengths of three sides of a triangle as inputs and outputs whether or not the triangle is a right triangle. (3

3 Differentiate between lists and tuples with the help of examples. (3

4 Write a Python program to print all palindromes in a line of text. (3

Module I

11 a) Discuss the steps involved in the waterfall model of software development process with the help of a neat diagram. (8)

b) Write a Python program to print all numbers between 100 and 1000 whose sum of digits is divisible by 9. (6)

OR

12 a) Illustrate the use of range() in Python. (6)

b) Write a Python program to print all prime factors of a given number. (8)

Module II

13 a) Write a Python program to compute the sum of the series ($1 - x^2/2! + x^4/4! - x^6/6! + \ldots\ldots\ldots$ n terms). (6)

b) Illustrate the use of any four dictionary methods. (8)

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OR

14 a) Write a Python program to convert a decimal number to its binary equivalent. (7)

b) Write a Python program to read a text file and store the count of occurrences of each character in a dictionary. (7)

PART A

Answer all questions, each carries 3 marks. Marks

1 Differentiate between definite iteration and indefinite iteration in Python (3)

2 Write a python program to reverse a number and also find the sum of the digits of that number. (3)

3 Define recursion and write a program in python to generate Fibonacci series using recursion. (3)

4 Define higher order functions used for mapping, filtering and reducing with examples. (3)

Module I

11 a) Discuss the steps involved in the waterfall model of software development process with the help of a neat diagram. (8)

b) Write a python program to print the pattern using for loop. (6)

```
5 4 3 2 1
4 3 2 1
3 2 1
2 1
1
```

OR

12 a) Write a Python program to print all numbers between 100 and 1000 whose sum (8)

of digits is divisible by 3.

b) Explain the concept of lazy evaluation in python with a programming example. (6)

Module II

13 a) Define the concept of Caesar cipher and write the programs for encryption and decryption with distance +3 for the lowercase alphabet. (8)

b) Define how dictionary associates data values and keys in python. Explain with syntax and example. (6)

a. Adding a key value pair.

b. Accessing values.

c. Traversing a Dictionary.

OR

14 a) Demonstrate the use of any four list methods with programming examples. (8)

b) Write a Python program to convert a decimal number to its binary equivalent. (6)

PART A

Answer all questions, each carries 3 marks.

1 What is the output of the following Python code. Justify your answer.

```
x = 'abcd'
for i in range(len(x)):
    print(i)
```

2 Write the syntax and semantics of the *multiway-if* statement.

3 Explain the concepts namespace, scope, and lifetime in the case of Python programming language.

4 What are mutable and immutable properties in the case of Python data structures? Give one example each for mutable and immutable data structures in Python.

Module I

11 a) Write a Python program to find the roots of a quadratic equation, $\mathbf{ax^2 + bx + c = 0}$. Consider the cases of both real and imaginary roots. (7)

b) Write a Python program to check whether a number is Armstrong number or not. (7)

An Armstrong number is an n-digit number that is equal to the sum of the nth powers of its digits.

Examples: $153 = 1^3 + 5^3 + 3^3$, $1634 = 1^4 + 6^4 + 3^4 + 4^4$

OR

12 a) Write a Python program to display the sum of odd numbers between a programmer specified upper and lower limit. (7)

b) (7)

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \ldots$$

Given the value of x, write a Python program to evaluate the following series upto n terms:

Module II

13 a) Write a Python program to implement Caesar cipher encryption and decryption on a string of lowercase letters. Take distance value and the string as input. (Hint: Caesar cipher encryption strategy replaces each character in the plaintext with the character that occurs a given distance away in the sequence. (7)

Encryption:Eg. input: 3, "invade", Eg. output: "lqydgh"

Decryption: Eg. input: 3, "lqydgh", Eg. output: "invade")

b) Write a Python code segment that opens a file for input and prints the number of four-letter words in the file. (7)

OR

14 a) Write a Python program to create a set of functions that compute the mean, median and mode of a set of numbers. Each function should expect a list of numbers as an argument and return a single number. Each function should return 0 if the list is empty. Include a main function that tests the three functions with a given list. (7)

(Hint: Mean: Mean is the average value of a list of numbers.

Median: If the number of values in a list is odd, the median of the list is the value at the midpoint when the set of numbers is sorted; otherwise, the median is the average of the two values surrounding the midpoint.

Mode: The mode of a list of values is the value that occurs most frequently)

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b) Write a Python program to check whether a list contains a sublist. (7)

Eg. Input 1: my_list = [3,4,5,2,7,8] , sub_list = [2,7]

output 1: True

input 2: my_list = [3,4,5,2,7,8] , sub_list = [5,7]

output 2: False

Module III

PART A

Answer all questions, each carries 3 marks. Marks

1 Jack says that he will not bother with analysis and design but proceed directly to coding his programs. Why is that not a good idea? (3)

2 Write the output of the following python statements : (3)

i) round(12.57) ii) 5//2 iii) int(6.5)

3 Write the output of following python code : (3)

```
S = "Computer"
print(S[::2])
print(S[::-1])
print(S[:])
```

4 Write a recursive function in python to find GCD of two numbers. (3)

Answer one full question from each module, each carries ...

Module I

11 a) Write a python program to find the sum of the cosine series 1 - x^2/2! + x^4/4!- ..... (7)

b) Write a python program to find X^Y or **pow(X,Y)** without using standard function (7)

OR

12 a) Write a python program to generate the following type of pattern for the given N rows where N <= 26. (7)

A

A B

A B C D

A B C D E

b) Write a python program to generate prime numbers within a certain range (7)

Module II

13 a) Assume that the variable data refers to the string "Python rules!". Use a string method to perform the following tasks: (6)

a. Obtain a list of the words in the string.

b. Convert the string to uppercase.

c. Locate the position of the string "rules" .

d. Replace the exclamation point with a question mark.

b) Write a code segment that opens a file for input and prints the number of four-letter words in the file (8)

OR

14 a) Assume that there is a text file named "numbers.txt". Write a python program to find the median of list of numbers in the file without using standard function for median. (10)

b) Use higher order python function **filter** to extract a list of positive numbers from a given list of numbers. You should use a lambda to create the auxiliary function. (4)